15-kW 800 V–48 V Dual Active Bridge Converter for AI Data Center Architectures

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***Abstract*—The rapid increase in power density of artificial intelligence (AI) data centers has created significant challenges in electrical power delivery, including increased conduction losses, scalability limitations, and thermal management constraints. To address these challenges, higher-voltage DC distribution architectures, such as 800 V DC systems, have been proposed to reduce current levels and improve overall efficiency. However, existing server and rack-level infrastructure continues to operate on a 48 V DC bus, creating a critical need for an efficient and scalable intermediate conversion stage.**

This paper presents the analytical design of an 800 V to 48 V isolated power conversion stage intended for rack-level integration within next-generation data center architectures. The proposed system consists of a front-end AC/DC rectification stage that establishes an 800 V DC bus, followed by a high-frequency isolated DC/DC conversion stage. A silicon carbide (SiC)-based full-bridge inverter generates a high-frequency AC waveform, which is transferred through a high-frequency transformer providing both voltage scaling and galvanic isolation.

On the secondary side, the stepped-down waveform is converted into a regulated 48 V DC output using a rectification stage, with emphasis on the trade-offs between diode-based and active rectification approaches under high-current conditions. The design framework highlights key considerations including transformer turns ratio selection, current stress on the low-voltage side, switching frequency impact on power density, and loss mechanisms in semiconductor and magnetic components.

The purpose of this work is to develop a scalable and efficient conversion interface that enables the integration of emerging 800 V DC distribution systems with established 48 V rack-level power delivery, supporting the continued growth of high-performance AI data center infrastructure.

***Index Terms*—800 V DC architecture, AI data centers, isolated DC–DC converter, high-frequency transformer, silicon carbide (SiC), full-bridge converter, rectifier design, rack-level power systems**

I. Introduction

The rapid expansion of artificial intelligence (AI) and high-performance computing workloads has significantly increased power density in modern data centers, with rack-level power now exceeding 100 kW. As AI clusters continue to scale, electrical power delivery has become a major system-level challenge due to increased conduction losses, thermal stress, and infrastructure complexity. Resistive conduction losses in electrical distribution are governed by

$$P_{loss} = I^2 R, \tag{1}$$

where I is the conductor current and R is the effective resistance of the distribution path. Because of the quadratic dependence on current, low-voltage, high-current architectures become increasingly inefficient at high power levels.

Delivered electrical power is defined by

$$P = VI, \tag{2}$$

where V is voltage and I is current. For a fixed power level, increasing the distribution voltage reduces current proportionally, leading to a significant reduction in conduction losses. This motivates the transition from conventional 400 V-class systems toward 800 V DC architectures for next-generation data centers.

Despite the efficiency benefits of high-voltage distribution, server and rack-level infrastructure remains largely standardized around a 48 V DC bus due to safety considerations, established power delivery ecosystems, and compatibility with existing rack and server architectures. Consequently, an efficient intermediate conversion stage is required to interface the high-voltage distribution bus with the low-voltage rack bus.

This transition is particularly relevant in AI-focused infrastructure developed by companies such as NVIDIA and its ecosystem partners. Modern AI platforms, including systems such as the NVIDIA GB200 NVL72, are representative of a broader industry shift toward extremely high rack-level power demand driven by dense GPU deployment, high-bandwidth interconnects, and sustained accelerator utilization. These workloads place substantial stress on traditional AC and low-voltage DC distribution systems, motivating the adoption of higher-voltage DC architectures to improve efficiency and scalability.

Fig. 1 illustrates a representative 800 V DC power delivery architecture aligned with NVIDIA-oriented AI deployments. As shown in Fig. 1, power is supplied from the utility grid and converted to a high-voltage DC bus through an active front-end rectifier stage. This 800 V DC bus is supported by energy buffering elements such as capacitor banks, battery energy storage systems (BESS), or uninterruptible power systems (UPS), which provide ride-through capability and

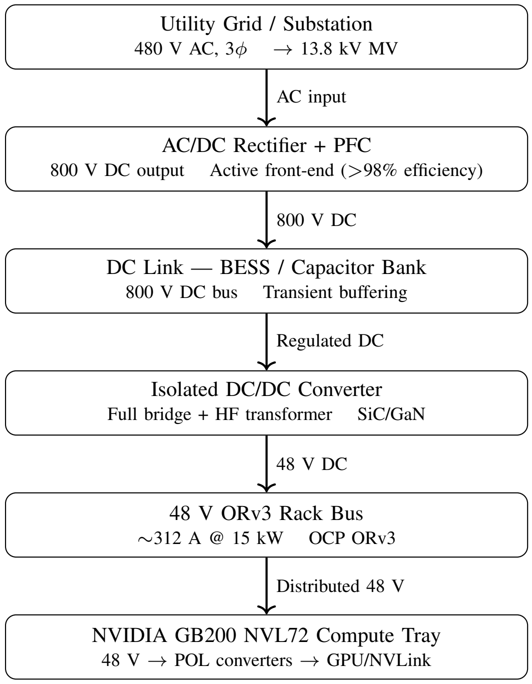

Fig. 1. Representative 800 V DC AI data center power architecture highlighting the transition from utility input to rack-level 48 V distribution and the isolated DC/DC conversion stage.

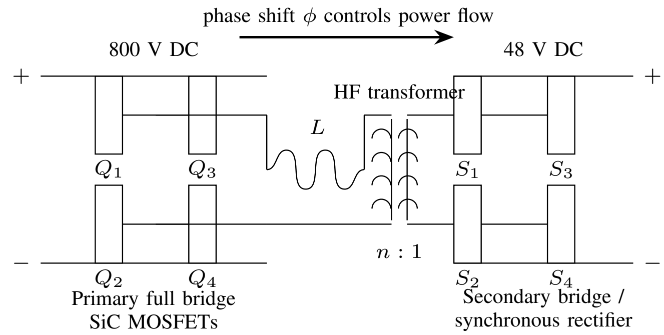

Fig. 2. Simplified dual active bridge (DAB) topology used for 800 V to 48 V isolated conversion. Power transfer is regulated through the phase shift between the primary and secondary bridge voltages.

transient load support. At the rack boundary, an isolated DC/DC conversion stage steps the 800 V DC bus down to a standardized 48 V distribution level compatible with Open Compute Project (OCP) ORv3 rack infrastructure. The 48 V bus then feeds point-of-load (POL) converters within GPU trays, where voltage is further regulated to the low-voltage levels required by GPUs, memory subsystems, and high-speed interconnects such as NVLink.

This system-level view motivates the converter-focused analysis developed in the remainder of the paper. After establishing the data center context in Fig. 1, the paper next introduces the proposed dual active bridge power stage in Fig. 2, then examines its power transfer behavior in Fig. 4, current stress in Fig. 5, magnetic design constraints in Fig. 6, and closed-loop regulation concept in Fig. 7.

II. System Architecture and Operating Principle

A. Overall System Structure

Rather than representing the converter as a sequence of abstract functional blocks, the rack-interface converter is more clearly described using the physical dual active bridge (DAB) structure shown in Fig. 2. As illustrated in Fig. 2, the converter consists of a high-voltage full bridge on the 800 V side, an effective series inductance that governs current slew and power transfer, a high-frequency transformer that provides galvanic isolation and voltage scaling, and a low-voltage full bridge or synchronous rectification stage on the 48 V side. Power transfer is controlled through the phase shift between the primary- and secondary-side bridge voltages.

Fig. 2 is especially important because it moves the discussion from the system architecture of Fig. 1 to the actual converter topology analyzed in the rest of the paper. Unlike a generic block diagram, this figure directly identifies the elements that determine converter behavior: the switching bridges, the transformer turns ratio, and the effective series inductance. These elements govern current stress, transferred power, soft-switching feasibility, semiconductor loss, and magnetic design constraints.

For a fixed output power, the motivation for the 800 V distribution bus follows directly from

$$P = VI \tag{3}$$

which implies that higher bus voltage reduces current. Since conduction loss is proportional to

$$P_{loss} = I^2 R, \tag{4}$$

reducing current yields a quadratic reduction in distribution losses. This is a major reason why 800 V architectures are increasingly attractive for AI data center power delivery. In this context, Fig. 2 represents the key intermediate conversion stage that enables the transition from the high-voltage facility bus of Fig. 1 to the rack-level 48 V bus.

B. AC/DC Front-End and DC Link

The front-end stage converts three-phase AC power into a high-voltage DC bus suitable for downstream conversion. This stage can be implemented using either a passive diode bridge or an active front-end converter, depending on system requirements for power quality and controllability.

For a three-phase diode rectifier, the average DC output voltage is approximated by

$$V_{DC} \approx \frac{3\sqrt{2}}{\pi} V_{LL} \tag{5}$$

where V_{LL} is the line-to-line RMS voltage. This produces a pulsating DC waveform that requires filtering to achieve a stable DC output.

A DC link capacitor is placed at the rectifier output to smooth voltage ripple and provide energy buffering. The capacitor supplies current during intervals when the rectifier is not conducting, thereby maintaining a relatively constant DC voltage. The sizing of the DC link capacitor depends on allowable voltage ripple, load power, and switching dynamics.

The adoption of an 800 V DC bus is primarily motivated by loss reduction. Since conduction losses scale with the square of current, increasing the bus voltage reduces current and significantly improves efficiency. This is particularly important in AI data centers, where high power levels lead to substantial thermal and electrical stress on distribution infrastructure.

In addition, the DC link serves as a decoupling stage, isolating upstream grid dynamics from downstream high-frequency switching behavior. This improves system stability and allows independent optimization of each conversion stage.

C. High-Frequency Isolated DC/DC Conversion Stage

The converter topology introduced in Fig. 2 identifies the primary switching bridge, the effective series inductance, the high-frequency transformer, and the secondary bridge or synchronous rectification stage. While Fig. 2 emphasizes the circuit-level structure of the dual active bridge, the power-stage interpretation in Fig. 3 highlights how energy is processed through the converter from the 800 V DC input to the regulated 48 V output.

As shown in Fig. 3, the input DC-link capacitor C_{dc} supports a stiff 800 V input by reducing low-frequency ripple and buffering instantaneous power mismatch between the upstream rectifier and the switching stage. The primary full bridge, implemented using SiC MOSFETs, converts the DC bus into a high-frequency square-wave voltage. This waveform excites the effective series inductance and transformer, which together determine the instantaneous current trajectory and average transferred power.

The transformer shown in both Fig. 2 and Fig. 3 provides galvanic isolation and voltage scaling. A first-order turns-ratio estimate is

$$n = \frac{V_1}{V_2} \approx \frac{800}{48} \approx 16.67, \tag{6}$$

although the final implemented ratio may differ slightly depending on conduction drops, allowable phase shift range, and regulation margin. High-frequency operation allows the transformer size to be reduced according to

$$B_{max} = \frac{V}{4NAf_s}, \tag{7}$$

where B_{max} is the peak flux density, N is winding turns, A is effective core area, and f_s is switching frequency.

The effective series inductance L, shown explicitly in Fig. 2 and functionally in Fig. 3, includes transformer leakage inductance and, if needed, additional external inductance. This inductance is not merely parasitic; it is a central design element because it governs the current ramp rate, ripple magnitude, and transferred power.

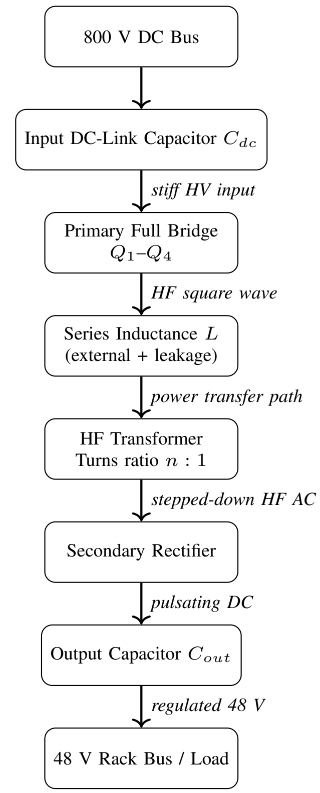

Fig. 3. Detailed power-stage topology of the proposed isolated 800 V to 48 V converter.

D. Secondary Rectification and Output Stage

The low-voltage output stage converts the stepped-down high-frequency AC waveform into a regulated 48 V DC output capable of supplying very large current to the rack bus. Two practical implementation paths are common: diode rectification and synchronous rectification.

Diode rectification offers simplicity, robustness, and minimal control complexity. However, the fixed forward voltage drop of the diodes becomes increasingly costly at high current. For the 15 kW, 48 V operating point considered here, the output current is

$$I_{out} = \frac{P}{V} = \frac{15000}{48} \approx 312.5 \text{ A}. \tag{8}$$

At this current level, even modest conduction drops translate into significant dissipation, which places considerable burden on thermal management and may reduce overall efficiency.

Synchronous rectification addresses this issue by replacing passive diodes with actively controlled MOSFETs. Because the effective conduction drop is determined by device on-state resistance rather than diode forward voltage, synchronous rectification can substantially reduce low-voltage-side conduction loss. The tradeoff is greater circuit complexity, tighter timing requirements, and added gate-drive and control considerations.

In either case, the output capacitor C_{out} is used to smooth the rectified waveform and maintain a stable low-ripple 48 V bus. Since the low-voltage side carries the highest current in

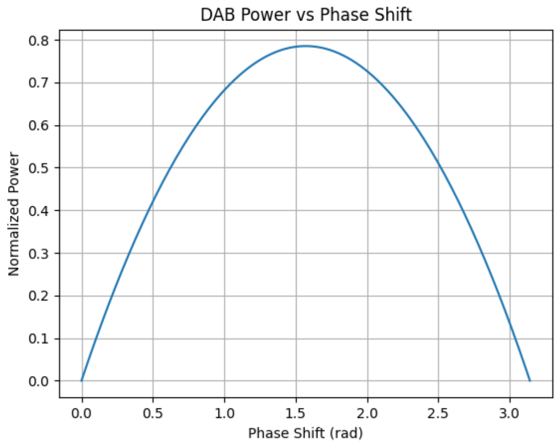

Fig. 4. Normalized transferred power versus phase shift under single-phase-shift control. The curve illustrates the nonlinear dependence of DAB power transfer on ϕ.

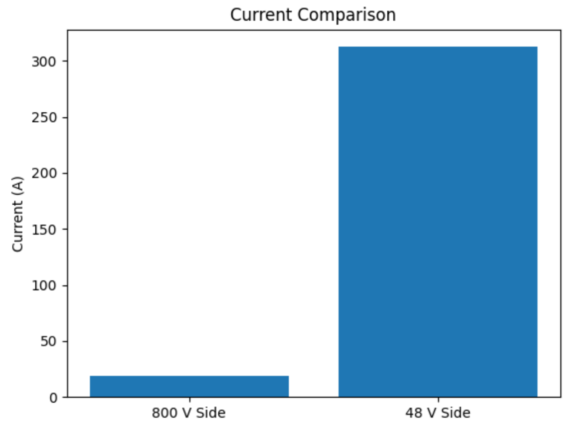

Fig. 5. Comparison of nominal current levels on the 800 V and 48 V sides for a 15 kW converter. The low-voltage side dominates conduction stress and thermal design.

the system, the output stage strongly influences packaging, thermal design, and achievable efficiency.

III. Analytical Power Transfer Modeling

This section develops an analytical framework to describe power transfer, current stress, and loss mechanisms in the proposed isolated DC/DC conversion stage. The analysis is based on the dual active bridge topology of Fig. 2 operating under single phase-shift modulation, which enables controlled power flow through phase displacement between primary and secondary bridge voltages. The resulting nonlinear relationship between transferred power and phase shift is illustrated in Fig. 4.

A. Single Phase-Shift (SPS) Power Transfer

Referring to the converter structure shown in Fig. 2, power transfer is governed by the phase relationship between the primary and secondary bridge voltages. Under single phase-shift modulation, both bridges operate at 50% duty cycle and a phase displacement ϕ is applied between primary and secondary bridge voltages. The average transferred power is commonly approximated by

$$P = \frac{nV_1V_2}{\omega L}\,\phi\left(1 - \frac{|\phi|}{\pi}\right), \tag{9}$$

This expression provides direct design insight into how voltage levels, inductance, and phase shift influence power transfer capability. Increasing the voltage product V_1V_2 or turns ratio n increases transferable power, while increasing L or switching angular frequency ω reduces transferable power for a given ϕ.

For small phase shifts, the relationship can be approximated as nearly linear:

$$P \approx \frac{nV_1V_2}{\omega L}\,\phi \qquad (\phi \ll \pi), \tag{10}$$

which is useful for initial control-oriented reasoning.

B. Turns Ratio Selection

A first-order turns ratio selection is based on the DC bus ratio:

$$n \approx \frac{V_1}{V_2} = \frac{800}{48} \approx 16.67. \tag{11}$$

This approximation ensures that the reflected voltage levels are reasonably matched, reducing unnecessary circulating current.

IV. Current Stress Analysis

A. Nominal Current Levels

For a 15 kW module, nominal currents can be estimated directly from the DC bus voltages:

$$I_{HV} = \frac{P}{V_1} = \frac{15000}{800} = 18.75 \text{ A}. \tag{12}$$

This result shows that the high-voltage side current remains relatively moderate, which reduces conductor loss and simplifies device current rating requirements on the 800 V side. On the 48 V side, however, the nominal current is significantly larger:

$$I_{LV} = \frac{P}{V_2} = \frac{15000}{48} \approx 312.5 \text{ A}. \tag{13}$$

This high current is a defining constraint of 48 V rack buses. The contrast between these two operating domains is summarized visually in Fig. 5, which highlights how strongly the low-voltage side dominates conduction stress.

B. RMS Current and Ripple Estimation

A first-order estimate of ripple current under SPS control can be written as

$$\Delta I \approx \frac{V_{eq}}{L}\cdot\frac{\phi}{\omega}, \tag{14}$$

TABLE I
DESIGN SPECIFICATIONS OF THE PROPOSED 15-KW DAB CONVERTER

Parameter	Value
Input voltage	800 V
Output voltage	48 V
Rated power	15 kW
Nominal HV current	18.75 A
Nominal LV current	312.5 A
Initial turns ratio estimate	16.67:1
Switching frequency	100 kHz (design target)
Control method	Single phase shift (SPS)

TABLE II
FIRST-ORDER ESTIMATED LOSS BREAKDOWN FOR THE PROPOSED CONVERTER

Loss component	Estimated value
Primary bridge conduction loss	120 W
Primary bridge switching loss	80 W
Transformer core loss	50 W
Transformer copper loss	100 W
Auxiliary / gate-drive loss	15 W
Total estimated loss	365 W
Estimated efficiency	97.6%

where V_{eq} represents the effective voltage difference driving the inductor during the phase-shift interval.

V. SEMICONDUCTOR LOSS MECHANISMS

A. Conduction Loss

Conduction loss in a MOSFET can be approximated by

$$P_{cond} = I_{rms}^2 R_{DS(on)}, \tag{15}$$

where $R_{DS(on)}$ is the on-state resistance.

B. Switching Loss

A first-order switching loss approximation using output capacitance is

$$P_{sw} \approx \frac{1}{2} C_{oss} V^2 f_s, \tag{16}$$

where C_{oss} is the device output capacitance and V is the switched voltage.

C. Soft-Switching Considerations (ZVS Feasibility)

Soft-switching is an important objective for high-frequency, high-density converters. A commonly used energy-based feasibility condition for zero-voltage switching compares the energy stored in the series inductance to the energy required to charge and discharge device capacitances:

$$\frac{1}{2} L I^2 \geq \frac{1}{2} C_{oss} V^2. \tag{17}$$

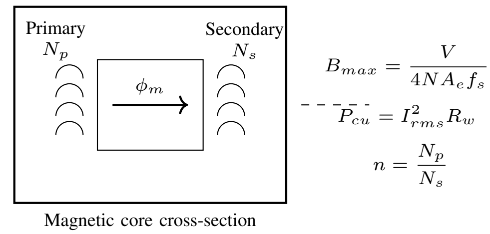

Fig. 6. Transformer design variables for the isolated DAB stage.

VI. TRANSFORMER AND MAGNETICS DESIGN CONSIDERATIONS

A. Flux Density Constraint

Peak flux density is often estimated using a square-wave excitation relationship:

$$B_{max} = \frac{V}{4NA_e f_s}, \tag{18}$$

where V is the applied winding voltage, N is the number of turns, A_e is the effective core area, and f_s is the switching frequency.

B. Copper Loss

Transformer winding copper loss can be approximated as

$$P_{cu} = I_{rms}^2 R_w, \tag{19}$$

where R_w is the winding resistance, including AC effects.

C. Skin Depth Consideration

Skin depth provides a first-order estimate of current penetration depth in conductors:

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}}, \tag{20}$$

where ρ is resistivity, μ is permeability, and ω corresponds to the operating frequency.

VII. CONTROL STRATEGY FORMULATION

A. Phase-Shift Control Concept

In the DAB converter, power transfer is regulated through the phase shift ϕ between bridge voltages. In a closed-loop control framework, ϕ serves as the manipulated variable used to regulate the 48 V output bus under varying load conditions.

A proportional-integral controller is commonly used, expressed in the Laplace domain as

$$\phi(s) = K_p e(s) + \frac{K_i}{s} e(s), \tag{21}$$

where K_p and K_i are proportional and integral gains.

The error signal is defined as

$$e(s) = V_{ref} - V_{out}. \tag{22}$$

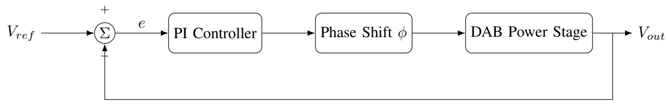

Fig. 7. Closed-loop phase-shift control for regulation of the 48 V output bus.

VIII. Research Challenges and Future Work

While the analytical framework above establishes a foundation, several research challenges remain for adapting DAB converters to AI data center environments. Key issues include minimizing circulating current under light load, extending the soft-switching region across wide operating conditions, managing high current density at 48 V, integrating transformer leakage inductance to reduce component count, and ensuring modular scalability and fault tolerance at the rack level.

These challenges motivate future work on advanced modulation strategies, refined loss modeling, and multi-objective optimization that jointly considers magnetics, semiconductor devices, thermal constraints, and control performance.

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